

Plant Disease Detection and AI-Powered Treatment Recommendation

Project Overview

This project is an AI-powered web application to detect plant diseases from leaf images and provides treatment recommendations. A custom Convolutional Neural Network (CNN) is used for image classification, while a Hugging Face Large Language Model (LLM) - Qwen2.5-7B-Instruct, generates detailed information about the detected disease, including symptoms, causes, treatment, and prevention. The application is built with PyTorch and deployed using Streamlit.

Problem Statement

Plant diseases can significantly reduce crop quality and yield. Identifying diseases early often requires expert knowledge, which may not always be available. This project aims to provide a simple tool that can identify plant diseases from images and generate understandable treatment recommendations using AI.

Project Objective

1. Build a CNN model for plant disease classification.
2. Compare Custom CNN model with Transfer-Learning models.
3. Integrate a Hugging Face LLM to generate treatment recommendations.
4. Deploy the solution as an interactive Streamlit application.

Dataset

This project uses a Kaggle dataset (dataset name - PlantVillage), which contains healthy and diseased plant leaf images for different crop species.

Dataset Details:

- Link : <https://www.kaggle.com/datasets/vipooooool/new-plant-diseases-dataset>
- Total Classes : 38
- Image Type : RGB
- Image Size : 256x256

Dataset Structure :

- Train Folder consists of 70295 images.
- Valid Folder consists of 17572 images.
- Test Folder consists of 33 Labeled images.

Data Preprocessing

All images were resized to **224 x 224** pixels before training.

Training images were augmented with random transformations to improve the model's ability to generalize. Validation images were only resized and normalized to ensure consistent evaluation.

The processed images were loaded using PyTorch [DataLoader](#) objects for efficient batch training.

Train Transforms :

Step	Purpose
Resize(224 x 224)	Standard image dimensions
Random Rotation(15)	To make model more robust
Random Horizontal Flip	Improves Generalization
Random Vertical Flip	Improves Generalization
Color Jitter (15)	For varying lighting conditions
To Tensor ()	To convert the image pixels into Tensor

Valid and Test Transforms :

Step	Purpose
Resize(224 x 224)	Standard image dimensions
To Tensor ()	To convert the image pixels into Tensor

Model Development

A Custom Convolutional Neural Network (CNN) model is developed using three Convolutional blocks followed by fully connected layers consisting of two hidden layers for classification.

Architecture :

Input Image (3 × 224 × 224)



Feature Extraction

Conv2D (3 → 16, Kernel=3×3, Padding='same')

ReLU

BatchNorm2D

MaxPool2D (2×2)

Conv2D (16 → 32, Kernel=3×3, Padding='same')

ReLU

BatchNorm2D

MaxPool2D (2×2)

Conv2D (32 → 64, Kernel=3×3, Padding='same')

ReLU

BatchNorm2D

MaxPool2D (2×2)

Dropout (0.2)

Classifier

Flatten

Linear (50176 → 128)

BatchNorm1D

ReLU

Dropout (0.3)

Linear (128 → 64)

BatchNorm1D

ReLU

Dropout (0.4)

Linear (64 → 38)

Output

38 Plant Disease Classes

Softmax (during inference)

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Architecture Summary :

Component	Details
Input Size	224 x 224 x 3
Convolution Layers	3
Batch Normalization	yes
Max Pooling	3
Dropout	0.2,0.3,0.4
Fully Connected Layers	3 (Including Output Layer)
Output Classes	38

Train Configuration :

Parameter	Value
Framework	PyTorch
Optimizer	Adam
Loss Function	CrossEntropy
Learning Rate	0.05
Epochs	30
Batch Size	32

Transfer Learning

Three pre-trained models were evaluated to compare their performance with the custom CNN.

- ResNet50
- EffecientNet_B0
- MobileNet_V02

Each model was fine-tuned by replacing the classification layer with a custom classifier for 38 output classes.

Model Evaluation

Each model is evaluated using the validation dataset.

Metrics used:

- Accuracy
- Recall
- Precision
- F1 Score
- Per-Class Accuracy
- Mis-Classified Samples

Model Comparison

All models were evaluated on the validation dataset after training. The custom CNN was trained for 30 epochs, while the transfer learning models (ResNet50, EfficientNetB0, and MobileNetV2) were fine-tuned for 10 epochs. The table below summarizes the evaluation metrics for each model.

Model	Epochs	Accuracy	Precision	Recall	F1 Score
Custom CNN	30	96.1%	96.2%	96.1%	0.960
ResNet50	10	96.5%	96.6%	96.5%	0.965
Efficient-NetB0	10	95.6%	95.7%	95.6%	0.959
MobileNet V2	10	95.9%	96.0%	95.9%	0.960

Observation: ResNet50 achieved the highest overall classification performance with an accuracy of **96.5%**. However, the custom CNN achieved comparable performance while requiring fewer computational resources, making it the preferred model for deployment.

AI Recommendation System

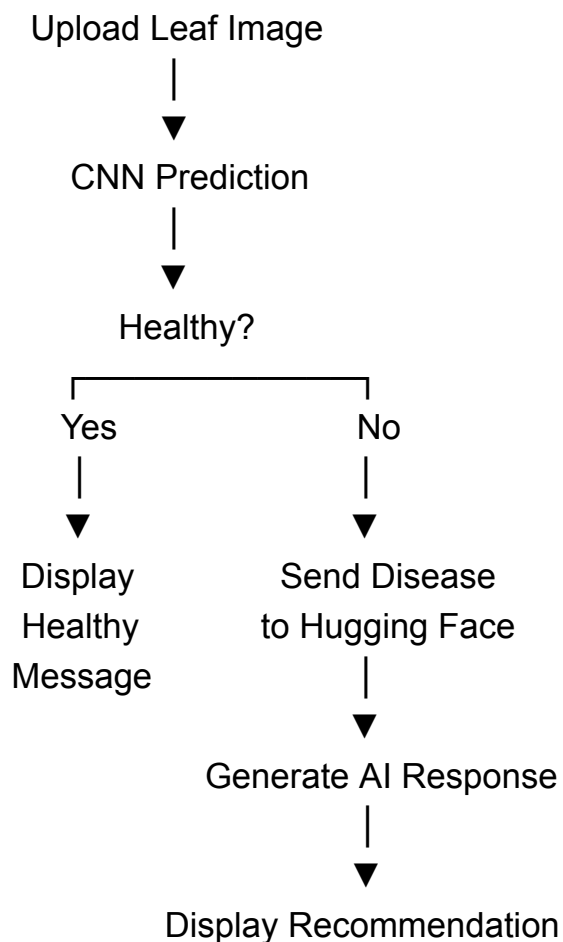
The classifier predicts only the disease name. To provide meaningful guidance, the predicted disease is sent to a Hugging Face Large Language Model **Qwen2.5-7B-Instruct** through the Inference API.

The model generates :

- Disease overview
- Symptoms
- Causes
- Treatment recommendations
- Prevention tips

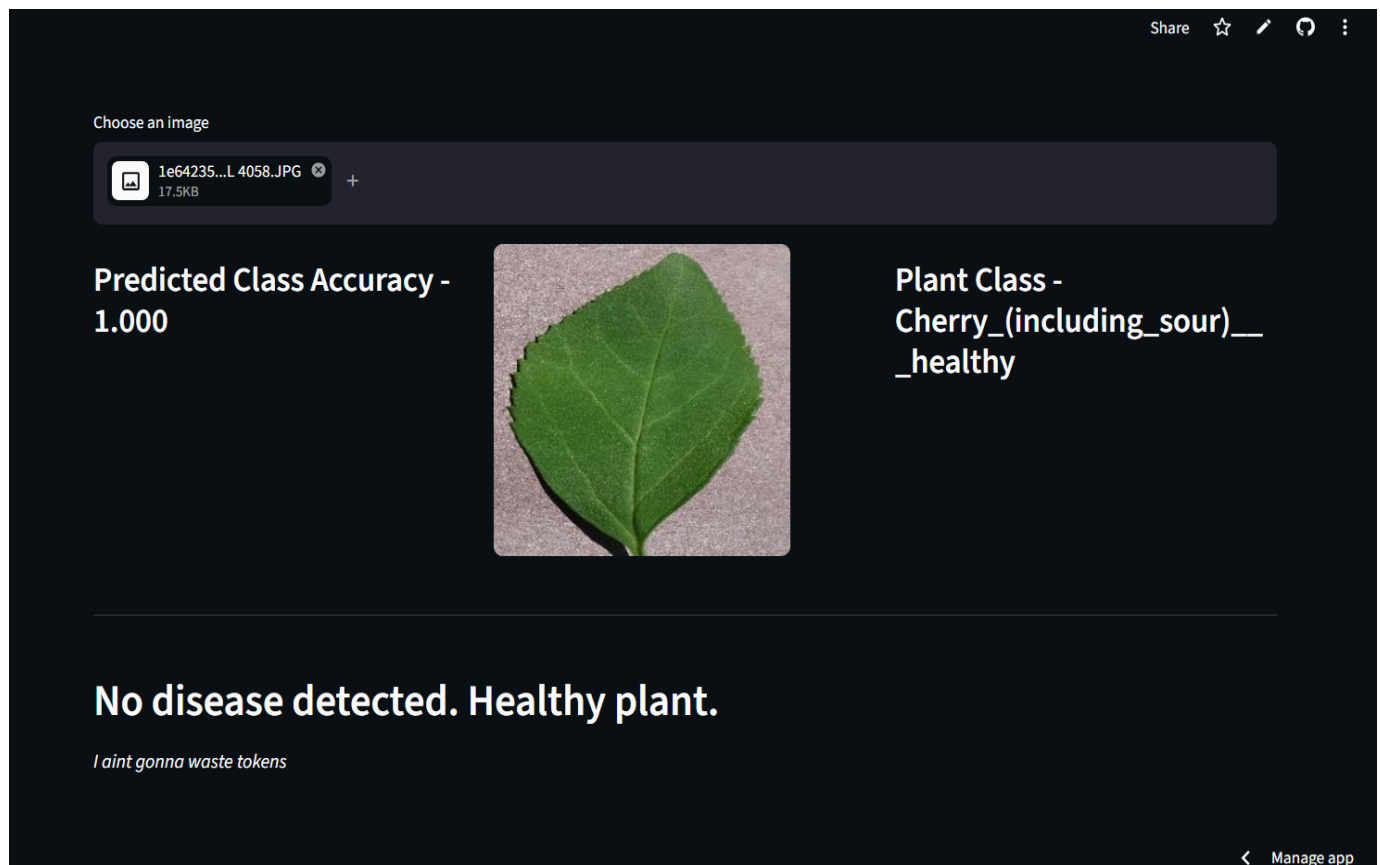
If the predicted class is healthy, the application skips the API call and displays a healthy status message instead to save some api tokens.

Application Workflow



Sample Output - HEALTHY PLANT

When a healthy plant is detected, the application displays the predicted class and confidence score. Since no disease is present, the Hugging Face API is not called, reducing unnecessary API requests.



Sample Output - DISEASED PLANT

When a diseased plant is detected, the predicted disease is sent to the Hugging Face Large Language Model, which generates a disease overview, symptoms, treatment recommendations, and prevention tips.

Choose an image

 23300cd...B 7103.JPG
15.5KB

Predicted Class Accuracy -
0.988



Plant Class -
Orange___Haunglongbing_
(Citrus_greening)

Treatment/Suggestion

Need Treatment Recommendations?

1. Disease Overview

Orange Huanglongbing, also known as Citrus Greening, is a serious bacterial disease that affects citrus trees. It's one of the most destructive diseases in citrus farming. The disease is spread by a tiny insect called the Asian citrus psyllid, which feeds on the leaves and stems of citrus trees.

2. Symptoms

- **Yellowing Leaves:** The leaves turn yellow, but the veins remain green, giving the leaves a mottled appearance.
- **Tissue Necrosis:** The fruit and leaves can develop brown spots or blotches.
- **Reduced Fruit Size and Quality:** Affected fruits are often smaller and less flavorful.
- **Tree Decline:** Over time, the tree may become weak and produce fewer fruits.

3. Causes

The disease is caused by a bacterium called *Candidatus Liberibacter asiaticus*. This bacterium is spread from tree to tree by the Asian citrus psyllid, which feeds on the sap of citrus trees. The psyllid can also spread the disease by moving from infected to healthy trees.

4. Treatment

Unfortunately, there is no cure for Huanglongbing. Once a tree is infected, it will eventually die. However, there are some steps you can take to manage the disease and slow its spread:

- **Remove Infected Trees:** If you find an infected tree, it's best to remove it to prevent the disease from spreading to other trees.
- **Monitor for Psyllids:** Regularly check your trees for the Asian citrus psyllid. If you find them, treat the trees with insecticides.
- **Use Resistant Varieties:** Some citrus varieties are more resistant to Huanglongbing. Consider planting these varieties if possible.

Deployment

The application is deployed using Streamlit Community Cloud. The trained CNN model is loaded during application startup, while disease recommendations are generated through the Hugging Face Inference API.

Although transfer learning models achieved slightly higher accuracy, the custom CNN was selected for deployment because it has a smaller model size, faster inference, and lower memory requirements while maintaining competitive performance.

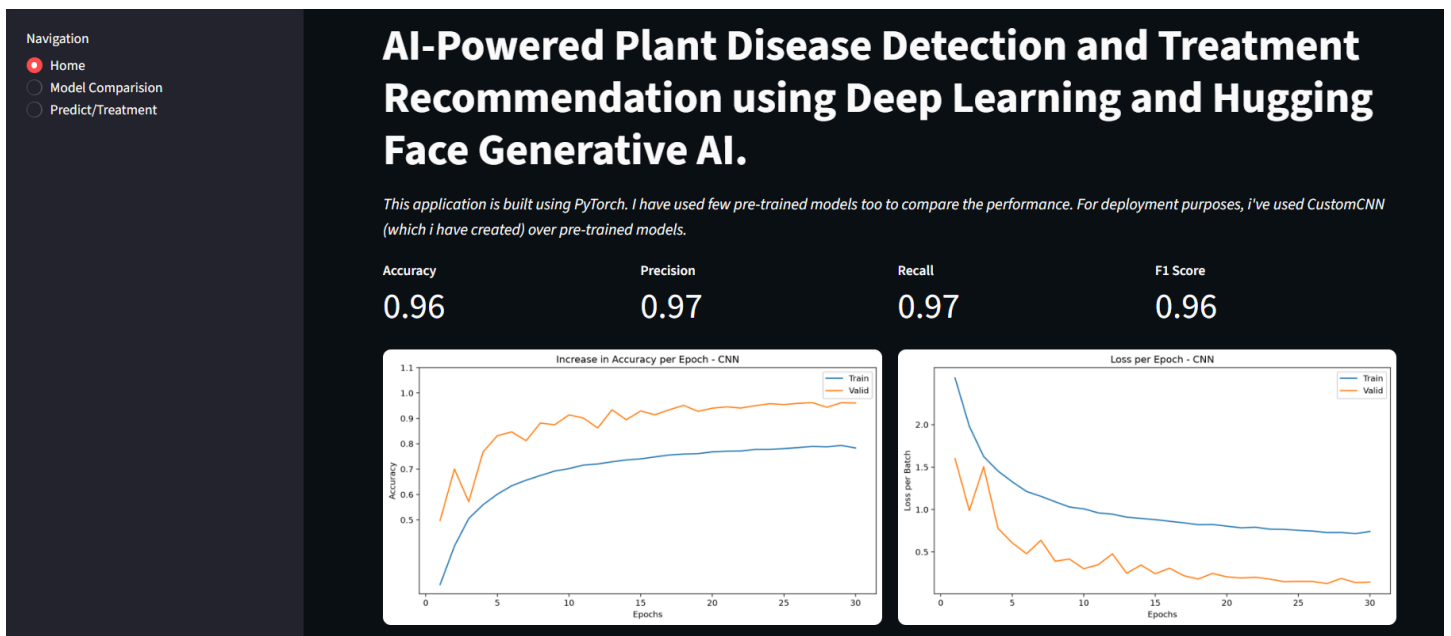
Link -

<https://plant-disease-detection-and-treatment-recommendation-bleh.streamlit.app/>

Results

The application successfully detects plant diseases from uploaded leaf images and provides AI-generated recommendations for diseased plants

The custom CNN achieved over **96% validation accuracy**, and the Hugging Face integration adds useful contextual information without increasing the complexity of the



classification model.

Limitations

- Limited to 38 classes in the PlantVillage dataset.
- Performance may vary on image quality, lightning or backgrounds.
- Limited tokens per month on LLM.

Conclusion

An end to end Deep Learning project to demonstrate a solution for plant disease detection and treatment recommendation generation. By combining a CNN-based image classified integrated with the Hugging Face Large Language Model. The application not only identifies plant diseases but also provides prevention as well as treatment guidance through an Steamlit Web-Application. An acceptable Computer Vision and Generative AI combined use case.